

FORM (03)

Graduation Project Proposal

1 Project Title

Design and Implementation of Virtualized Security Operation Center for Threat Simulation an...

2 Introduction

With a growing Cyber Threat Landscape , the integration of digitized information has risen the risks for organizations and end users alike . Hence operating without a dedicated Security Operations Center (SOC) increases incident response times and forces non-specialists IT staff to handle complex security events; Absence of Proactive Threat Hunting and structured compliance workflows . To bridge the gap between theoretical knowledge and practical application, this project focuses on building a virtualized SOC Environment. Such an environment provides a secure, economical framework to practice configuring, optimizing, and securing IT infrastructure . By simulating enterprise networks, we can actively analyze network data for anomalies and understand normal traffic patterns .

3 Brief Literature Review

Previous studies highlight the necessity of isolated testing environments for penetration testing, as well as automated response systems like CYBEX to rapidly publish defensive rules 1 . Furthermore, broad literature reviews underline the exponential growth of malware, phishing, and DDoS attacks, supporting the need for continuous threat monitoring using machine learning and big data pipelines 2 . However, many existing academic frameworks lack hands-on, hypervisor-based labs that replicate enterprise SOC orchestration. Our project addresses this by deploying an interconnected virtual environment combining open-source defensive tools.

ref 1 : DOI: 10.1109/ICCCNT56998.2023.10306937

ref 2 : DOI: 10.18535/ijssrm/v9i12.ec04

4 Methodology and project stages

Stage 1 / Defining core monitoring objectives and establish project boundaries
Stage 2 / Establishing baseline security components
Stage 3 / Developing the operational architectures
Stage 4 / Formalizing event loggin requirements
Stage 5 / Deploying an open-source securirty toolkit within the hypervisor
Stage 6 / Building a structured log management pipeline to collect, parse, and store endpoing telemetry
Stage 7 / Continuously find-tuning detection rules to minimize false postives

5 Time Plan for Each Stage (During the Semester)

Time Tasks	First Month				Second Month				Third Month			
	1	2	3	4	1	2	3	4	1	2	3	4
Stage 1 / Setting Targets												
Stage 2 / Implementing Core Functi...												
Stage 3 / Designing the SOC												
Stage 4 / Process, procedures and T...												
Stage 5 / Environment Preparation												
Stage 6 / Capability Deployment												
Stage 7 / Maintenance												

6 Tasks For Each Group Members

IBRAHIM: SIEM architecture and log pipeline orchestration. creating centralized event indexes, designing monitoring dashboards, and tuning correlation queries for telemetry analysis.

FAISAL: Network security monitoring and perimeter defense; installing firewall routing and subnet isolation, and deploying Security Onion (Suricata IDS/IPS) to capture live network traffic and detect packet anomalies.

SULTAN: Endpoint telemetry and directory infrastructure; setting up Windows Server, deploying Windows and Linux endpoints, and configuring Universal Forwarders to ship event logs to the SIEM.

Member 4 : Red team threat simulation and playbook validation; executing controlled attacks, verifying alert triggers, and documenting incident response playbooks

7 Expected Results

The primary outcome of this project will be a fully operational, virtualized Security Operations Center (SOC) capable of end-to-end event collection, threat detection, and real-time incident analysis. Key deliverables include:

Integrated Virtual SOC Environment: An isolated, four-segment virtual network (Attacker, Firewall, DMZ/Monitoring, and Protected Domain) running production-grade defensive tools including pfSense, Security Onion, and Splunk/Wazuh.

Automated Log Ingestion Pipeline: An established data collection path forwarding endpoint telemetry (via Sysmon and Universal Forwarders) and network packet data into a centralized indexer.

Detection Engineering Repository: A curated suite of detection rules and SIEM queries mapped to common adversary techniques, validated by controlled red team simulations.

Incident Response Playbooks & Documentation: Standard operating playbooks detailing triage, containment, and root-cause analysis for simulated threats, serving as a reproducible framework for practical blue-team training.

8 Signatures

☒ I acknowledge that I have read, understand and agree the above terms ...

SUPERVISORS (FACULTY MEMBERS)

S1 -	S2 -

STUDENTS (GROUP MEMBERS)

1 IBRAHIM AL-BAKER	2 FAISAL
3 SULTAN	4 -